

The equivalent translational compliance of steel or wood studs and resilient channel bars^{a)}

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A number of recent papers have determined the compliance of steel studs for use in models for predicting the sound insulation of cavity stud walls. However, in these papers, the compliance of resilient channel bars on one or both sides of wood studs or on one side of steel studs has not been determined across the whole of the frequency range. The present paper determines the compliance of the combination of resilient channel bars, mounted on wooden or steel studs and modeled as point or as line connections. Steel studs have usually been modeled as line connections. In this paper, they are also modeled as point connections where the points are the screws attaching the wall leaves to the steel studs and individual results rather than average results are analyzed. The compliance that makes Davy's sound insulation predictions agree with experimental sound insulation data was calculated by inverting Davy's equations for sound insulation. Linear regressions of the logarithm of the compliance, against the logarithms of frequency, reduced surface density, cavity depth and number of point connections or stud spacing, were conducted in a low frequency range and in a high frequency range. © 2015 Acoustical Society of America. [<http://dx.doi.org/10.1121/1.4916706>]

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I. INTRODUCTION

Models for the predicting the sound insulation of cavity stud walls were developed by number of researchers. Heckl (1959a,b) and Cremer *et al.* (2005a) developed prediction formulae for the sound power radiated on one side of an infinite plate excited by a point force and the sound power per unit length radiated from one side of an infinite plate excited by an infinite line source. Sharp (1973, 1978) and Sharp *et al.* (1980) developed the prediction of the sound insulation of lightweight cavity walls with rigid studs or rigid point connections by using Heckl's results.

A prediction method for a cavity wall with resilient steel studs was developed by Gu and Wang (1983). They modeled the resilient steel studs as springs with an equivalent translational stiffness. Davy (1990a,b) suggested that Gu and Wang's formulae "are not obviously an extension of Sharp's formulae" and introduced an equivalent mechanical compliance (the inverse of equivalent mechanical stiffness). Gu and Wang's formulae are incorrect because they used the impedance of the mass of the effective near field vibrational radiating area of the radiating panel rather than the actual point or line impedance of the radiating panel. All of these results only apply below the critical frequency of the wall. Davy (1991) extended his formula to above the critical frequency, and his research was supported by Hongisto (2006). Hongisto showed that Davy's theory agreed with

measurements on the steel stud walls with sound absorption in the cavity while Gu and Wang's theory did not.

Poblet-Puig *et al.* (2006) calculated the vibrational level difference of different gypsum plaster board wall leaves that were connected via steel studs. They compared these differences with those calculated for line connections with a range of equivalent translational stiffnesses and equivalent rotational stiffnesses. Guigou-Carter and Villot (2006) used this information to calculate the sound insulation at low frequencies of two gypsum plaster board steel stud cavity walls with sound absorbing material in the wall cavity. At higher frequencies, they modeled the steel studs as resilient point connections situated at the positions of the screws used to attach the gypsum plaster board to the steel studs.

Poblet-Puig *et al.* (2006, 2009) have shown that a steel stud can be modeled as a translational spring with an equivalent translational stiffness which varies with frequency. Vigran (2010) derived a best-fit third order polynomial approximation to the logarithm of Poblet-Puig's numerical values as a function of the logarithm of the frequency for the most common type of steel stud.

Davy *et al.* (2012) determined the equivalent translational compliance of steel studs when modeled as a line connection. The compliance of steel studs needs to be known to use a simple theoretical sound insulation prediction model of steel stud gypsum plaster board cavity walls.

Davy (2012) derived equations for the transmission factor τ of a cavity wall via resilient massless point and line connections. The only theoretical model for predicting the sound insulation of cavity stud walls with resilient channel bars that the authors are aware of is Bradley and Birta (2001). This models the connections as having constant compliance and thus is only valid at low frequencies. A general prediction model would avoid the need to scan through a large number of sound insulation measurements and thus

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would probably be widely used. It should be possible to develop a similar method for predicting the sound insulation of wall leaves with resilient channel bars by using the same method as Davy *et al.* (2012).

The aim of this paper was to produce equations for the effective translational compliance of wooden and steel studs with resilient channel bars and steel studs using line or point connection theories. The equivalent translational compliance was calculated from experimental sound insulation data by inverting the formulae of Davy's current sound insulation theory. Linear regressions of the logarithm of the compliance, against the logarithms of frequency, the reduced surface density of the wall leaves, the cavity depth and the number of point connections or the stud spacing, were conducted in the low and the high frequency ranges. The equations produced by these linear regressions can be used in existing sound insulation prediction models such as those of Davy (2012).

II. AIRBORNE TRANSMISSION ACROSS THE CAVITY ABOVE THE CRITICAL FREQUENCY

This section and the next two sections summarize the formulae of the current version of Davy's sound insulation theory that was used in this paper. This is done so that readers do not have to hunt through a number of Davy's papers.

For frequencies greater than or equal to the critical frequency, the maximum value of the transmission coefficient occurs at the coincidence angle of θ_{ci} . The angle θ_{ci} can be determined from the ratio of the critical frequency to the actual frequency

$$\sin^2 \theta_{ci} = \xi_i = \frac{f_{ci}}{f}. \quad (1)$$

The radiation efficiency can be determined using the following procedure. First, the cosine of the coincidence angle is calculated from Eq. (2). The cosine of the coincidence angle is set to zero for frequencies below the critical frequency.

$$g = \begin{cases} \cos \theta_{ci} = \sqrt{1 - f_{ci}/f} & \text{if } f \geq f_{ci} \\ 0 & \text{if } f < f_{ci}. \end{cases} \quad (2)$$

Second, the length of the side of an equivalent square panel is defined as

$$2l = \frac{4S}{U}. \quad (3)$$

S is the area of the panel and U is the perimeter of the panel.

The three constant terms used in the calculation of the radiation efficiency are $n=2$, $w=1.3$, $\beta=0.124$ (Davy, 2009a).

Third, some intermediate values are calculated using the wavenumber k

$$p = \begin{cases} w\sqrt{\frac{\pi}{2ka}} & \text{if } w\sqrt{\frac{\pi}{2ka}} \leq 1 \\ 1 & \text{if } w\sqrt{\frac{\pi}{2ka}} > 1, \end{cases} \quad (4)$$

$$h = \frac{1}{2/3\sqrt{2ka/\pi} - \beta}, \quad (5)$$

$$\gamma = \frac{h}{p} - 1, \quad (6)$$

$$q = \frac{2\pi}{k^2 S}. \quad (7)$$

Finally, the radiation efficiency is calculated as follows:

$$\sigma(\theta_{ci}) = \begin{cases} \frac{1}{\sqrt[n]{g^n + q^n}} & \text{if } 1 \geq g \geq p \\ \frac{1}{\sqrt[n]{(h - \gamma g)^n + q^n}} & \text{if } p > g \geq 0. \end{cases} \quad (8)$$

The diffuse field sound transmission coefficient is

$$\tau_d = \frac{I}{s_1^2 s_2^2 \alpha^2}, \quad (9)$$

where

$$s_i = \frac{2a_i \xi_i}{\sigma(\theta_{ci})}. \quad (10)$$

The $\sigma(\theta_{ci})$ is the single sided forced radiation efficiency at the coincidence angle of the i th wall leaf, and α is the sound absorption coefficient of the wall cavity of the double leaf wall.

I is

$$I = \frac{A + B + C}{D}, \quad (11)$$

where

$$P_i = 1 - \frac{1}{\xi_i}, \quad (12)$$

and ξ_i is the ratio of frequency to critical frequency of the i th leaf.

$$Q_i = \frac{1 + \frac{a_i \eta_i}{\sigma(\theta_{ci})}}{s_i}, \quad (13)$$

$$A = Q_1 Q_2 (P_2 - P_1) \ln \left\{ \frac{[Q_1^2 + (P_1 - 1)^2] (Q_2^2 + P_2^2)}{[Q_2^2 + (P_2 - 1)^2] (Q_1^2 + P_1^2)} \right\}, \quad (14)$$

$$B = Q_1 \left[(P_1 - P_2)^2 + Q_1^2 - Q_2^2 \right] \times \left[\arctan \left(\frac{P_2}{Q_2} \right) - \arctan \left(\frac{P_2 - 1}{Q_2} \right) \right], \quad (15)$$

$$C = Q_2 \left[(P_2 - P_1)^2 + Q_2^2 - Q_1^2 \right] \times \left[\arctan \left(\frac{P_1}{Q_1} \right) - \arctan \left(\frac{P_1 - 1}{Q_1} \right) \right], \quad (16)$$

$$D = Q_1 Q_2 [(P_2 - P_1)^2 + (Q_2 + Q_1)^2] [(P_2 - P_1)^2 + (Q_2 - Q_1)^2]. \quad (17)$$

If $Q_1 = Q_2 = Q$, $P_1 = P_2 = P$, the preceding Eqs. (11) to (17) become

$$I = \frac{Q^2 - P(P-1)}{2Q^2(Q^2 + P^2) [Q^2 + (P-1)^2]} + \frac{\arctan(P/Q) - \arctan((P-1)/Q)}{2Q^3}. \quad (18)$$

Davy (2010) stated that $\sigma(\theta_{ci})$ is modified by applying two conditions to obtain good agreement with experimental results. First, if the frequency is greater than or equal to the lower of the two critical frequencies, set $\sigma(\theta_{ci})$ equal to 1. Second, if the frequency is between 0.9 and 1 times the lower of the two critical frequencies, $\sigma(\theta_{ci})$ is linearly interpolated in the frequency domain between the values of $\sigma(\theta_{ci})$ at 0.9 and 1 times the lower of the two critical frequencies.

III. AIR BORNE TRANSMISSION ACROSS THE CAVITY BELOW THE CRITICAL FREQUENCY

The normal incidence mass-air-mass resonance frequency f_{mam} of a cavity wall is given by (Fahy and Gardonio, 2007)

$$f_{mam} = \frac{1}{2\pi} \sqrt{\frac{\rho_0 c^2 (m_1 + m_2)}{dm_1 m_2}}, \quad (19)$$

where ρ_0 is the density of air 1.18 kg m/s, c is the speed of sound in the air 343 m/s, m_i is the mass per unit area of the i th gypsum wall leaf ($i = 1$ or 2), and d is the distance between the two wall leaves across the wall cavity. The cavity wall behaves like a single wall with the same total mass per unit area below the normal incidence mass-air-mass resonance frequency.

Its diffuse field incidence sound transmission coefficient at certain frequency f when f is below f_{mam} is

$$\tau_f = \frac{1}{a^2} \ln \left(\frac{1 + a^2}{1 + a^2 \cos^2 \theta_l} \right), \quad (20)$$

where

$$a = \frac{f\pi(m_1 + m_2)}{\rho_0 c} = a_1 + a_2, \quad a_i = \frac{f\pi m_i}{\rho_0 c}, \quad (21)$$

where $\rho_0 c$ is the characteristic impedance of air.

An important fact is that Eq. (21) is a modified version of Eqs. (15) and (24) of Davy (2009b). This modified equation omits the correction term for below critical frequency proposed by Sewell (1970). The removal of Sewell's correction term in Eq. (21) for the sound transmission coefficient above the normal incidence mass-air-mass resonance frequency and below the upper of the two frequencies, is compensated for by calculating the sound transmission coefficient as the sum of Eqs. (9) and (22). At and above the upper of the two critical frequencies, only Eq. (9) is used.

Equation (22) can be included immediately above the upper of the two critical frequencies, but it causes problems if used a long way above both critical frequencies.

The diffuse field incidence sound transmission coefficient when $f_{mam} < f < f_{cl}$ is

$$\tau_f = \frac{1 - \cos^2 \theta_l}{(N + M \cos^2 \theta_l)(N + M)}, \quad (22)$$

where

$$N = \frac{1}{2} \left(\frac{a_2}{a_1} + \frac{a_1}{a_2} \right), \quad (23)$$

$$M = a_1 a_2 \alpha, \quad (24)$$

and the f_{cl} is the lowest of the critical frequencies of the two wall leaves.

Equation (22) is the Eq. (33) of Davy (2009b), and Eqs. (23) and (24) are Eqs. (29) and (30) of Davy (2009b). To avoid confusion with the symbols used earlier, p and q are replaced by M and N , respectively. N is half the ratio of the surface density of wall leaf two to the surface density of wall leaf one plus half the inverse of this ratio. If the wall leaves are the same, then $N = 1$. M is the product of the ratio of the mass impedance of one wall leaf to twice the specific acoustic impedance of air with the same ratio for the other wall leaf, and with the sound absorption coefficient of the cavity.

To obtain better agreement with experimental results and Sharp's theory for cavity walls with sound absorbing material in the cavity, θ_l needs to be limited to less than or equal to 61° (Davy, 2009b). Because θ_l has to be greater than zero for Eq. (22) to produce positive results, $\cos^2 \theta_l$ is limited to be less or equal to 0.9. Thus the condition of $\cos^2 \theta_l$ will be as follows:

$$\cos^2 \theta_l = \begin{cases} 0.9 & \text{if } \frac{1}{k\sqrt{A}} > 0.9 \\ \frac{1}{k\sqrt{A}} & \text{if } 0.9 \geq \frac{1}{k\sqrt{A}} \geq \cos^2 61^\circ \\ \cos^2 61^\circ & \text{if } \cos^2 61^\circ > \frac{1}{k\sqrt{A}}. \end{cases} \quad (25)$$

The middle line of Eq. (25) implements the radiation efficiency of the finite size wall radiating into an infinite half space. The last line of Eq. (25) is needed because the radiation efficiency into the wall cavity is different from the radiation efficiency into an infinite half space.

Because the absorption of any absorbing material in the cavity is limited by the depth of the cavity, the maximum value of the absorption coefficient α is also limited

$$\alpha \leq \begin{cases} kd & \text{if } kd < 1 \\ 1 & \text{if } kd \geq 1. \end{cases} \quad (26)$$

This limitation also makes the theory agree with Sharp's theory for cavity walls with sound absorbing material in the cavity. The absorption coefficient of a cavity with sound absorbing material in it is set equal to 1.

The absorption coefficient for an empty cavity is (Davy, 2010)

$$\alpha = 0.043 + 0.73d. \quad (27)$$

This equation was derived empirically from sound insulation data in Halliwell *et al.* (1998) and NAHB (1971).

For frequencies between 2/3 of the mass-air-mass resonant frequency and the mass-air-mass resonant frequency, the sound reduction index is interpolated in the logarithmic frequency domain using the sound reduction calculated from Eq. (20) at 2/3 of the mass-air-mass resonant frequency and the sound reduction index calculated from Eq. (22) at the mass-air-mass resonant frequency. The factor of 2/3 is derived from comparisons with experimental data.

IV. STUD BORNE TRANSMISSION ACROSS THE CAVITY

(Davy, 2012) derived equations for the sound transmission factor τ of a cavity wall via resilient massless point and line connections

$$\tau = \frac{16n\rho_0^2 c^4 QR}{\pi^5 f^2 [(m_1 f_{c2} + m_2 f_{c1})^2 + 64f^2 m_1^2 m_2^2 c^4 C^2]} \quad (\text{Point connection}), \quad (28)$$

$$\tau = \frac{8\rho_0^2 c^3 QR}{[g^2 + (8\pi f^{3/2} m_1 m_2 c C - g)^2] b \pi^3 f^2} \quad (\text{Line connection}). \quad (29)$$

f is the frequency, n is the number of point connections per unit area calculated from the stud spacing and the resilient channel bar or screw spacing, b is the distance between the line connections (stud spacing), ρ_0 is the density of air 1.18 kg/m³, c is the speed of sound in the air 343 m/s, m_i is the mass per unit area of the i th gypsum wall leaf, f_{ci} is the critical frequency of the i th wall leaf, and C is the compliance of the connection. The compliance C has dimensions of length/(force/length) = 1/pressure for a line connection, and length/force for a point connection.

Q is

$$Q = \begin{cases} 1 + e & \text{if } f < f_{c1} \\ e & \text{if } f \geq f_{c1}. \end{cases} \quad (30)$$

e is

$$e = \frac{\pi f_{c1} \sigma_1}{4f \eta_1}. \quad (31)$$

According to Cremer *et al.* (2005b), e is the ratio of the resonant vibrational energy of the first wall to its mass law vibrational energy, σ_i is the radiation efficiency, and η_i is the total damping loss factor of the i th wall leaf, respectively. Davy (2010) limits the maximum value of the radiation efficiency to 1. The condition for Q needs to be stated because above the critical frequency it is not possible to distinguish between the forced response given by the mass law response, and the resonant response at and above the critical frequency.

R is calculated by using following equation:

$$R = \begin{cases} 1 + r & \text{if } f < f_{c2} \\ r & \text{if } f \geq f_{c2}, \end{cases} \quad (32)$$

where r is the ratio of the power radiated by the resonant velocity of the second leaf and the power radiated by the vibrational near field in the second wall leaf generated by the connection

$$r = \frac{\pi f_{c2} \sigma_2}{4f \eta_2} \quad (\text{Point connection}), \quad (33)$$

$$r = \frac{\sigma_2}{2\eta_2} \sqrt{\frac{f_{c2}}{f}} \quad (\text{Line connection}). \quad (34)$$

The value of r is dependent on the connection type and thus so is R (Davy, 2012).

Because the line connection equation is non-symmetrical, the wall leaves need to be numbered so that $f_{c1} < f_{c2}$. This is because total internal reflection will occur for some propagation directions if the wall leaves are numbered in the opposite order.

g is

$$g = m_1 \sqrt{f_{c2}} + m_2 \sqrt{f_{c1}}. \quad (35)$$

The reduced surface density can be calculated from the following equation:

$$m_r = \frac{m_1 m_2}{m_1 + m_2}. \quad (36)$$

m_i is the mass per unit area of the i th gypsum wall leaf. The variable m_r is used because it is how the two surface densities are combined in the equation used to calculate normal incidence the mass-air mass resonant frequency of the cavity wall. Equations (28) and (29) can be simplified to the following equations when the properties of the two walls are the same:

$$\tau = \frac{4n\rho_0^2 c^4 Q}{\pi f^2 m^2 [f_c^2 + 16f^2 m^2 c^4 C^2]} \quad (\text{Point connection}), \quad (37)$$

$$\tau = \frac{2\rho_0^2 c^3 QR}{[f_c + (4\pi f^{3/2} m c C - \sqrt{f_c})^2] b \pi^3 m^2 f^2} \quad (\text{Line connection}). \quad (38)$$

Equations (28) and (29) will be combined with the theory for airborne sound transmission across the cavity presented in the last section. The stud borne transmission is included in the combined transmission below the mass-air-mass resonance frequency. This is different from Davy (2010) in which the stud borne transmission is not included below the mass-air-mass resonance frequency.

V. COMPLIANCE CALCULATIONS OF LINE AND POINT CONNECTIONS

The empirical equivalent translational compliance was the value that made the theoretical and experimental values of the sound reduction index agree with each other. The empirical equivalent translational compliances obtained by Davy *et al.* (2012) were calculated by trial and error. For this paper, the equations were inverted.

The equations for calculating the equivalent translational compliance are

$$\tau_s = \tau_t - \tau_a, \quad (39)$$

$$\tau_t = 10^{-(R_t/10)}, \quad (40)$$

$$\tau_a = 10^{-(R_a/10)}, \quad (41)$$

where τ_s is the structure borne transmission coefficient, τ_t is the total transmission coefficient given by the experimental data, τ_a is the calculated theoretical air borne transmission coefficient, R_t is the measured total sound reduction index, and R_a is the calculated air borne sound reduction index.

The calculation for the compliance is obtaining by inverting Eqs. (28) and (29)

$$C_M = \sqrt{\frac{(16n\rho_0^2 c^4 QR)/\pi^5 f^2 \tau_s - (m_1 f_{c2} m_2 f_{c1})}{64f^2 m_1^2 m_2^2 c^4}} \quad (\text{Point connection}), \quad (42)$$

$$C_M = \frac{\sqrt{8c^3 \rho_0^2 QR/\pi^3 \tau_s b f^2 - g^2 + g}}{8\pi c f^{3/2} m_1 m_2} \quad (\text{Line connection}). \quad (43)$$

The compliance must be a real non-negative number. These inverted equations do not return a real value when the expression under the square root is negative. This occurs in two situations. First is when the theoretical sound transmission via the air in the wall cavity is higher than the total experimental value. Second is when the sum of the theoretical sound transmission via the air in the cavity and via rigid structural connections is lower than the experimental value. In the first case, the best agreement with experiment is obtained by setting the theoretical compliance to the infinity. In the second case, the best agreement with experiment is obtained by setting the theoretical compliance to zero.

VI. PARAMETER CALCULATION

Davy *et al.* (2012) assumed that the volume density and Young's modulus of the gypsum plaster board were generic constants. This paper used the actual surface densities and the actual Young's moduli of the different gypsum plaster boards.

The density ρ_i is given by

$$\rho_i = \frac{m_i}{d_i}. \quad (44)$$

m_i and d_i are the surface density and thickness of the gypsum plasterboard of the i th wall leaf.

The Young's moduli used in the calculations are calculated from measurements of the flexural stiffness. However, the orientation of the gypsum plaster board sheets is not known. Therefore the Young's moduli used are the average of the gypsum plaster board values along and across the sheet. Table I shows the averaged Young's moduli used in the calculations for each type of gypsum plaster board.

The *in situ* damping loss factor of gypsum plaster board is assumed to be 0.03 because this value gave good agreement between predicted and experimental sound insulation values for double stud gypsum plaster board walls with sound absorbing material in their cavities (Davy *et al.*, 2012). The *in situ* damping loss factor has a significant effect on the theory for airborne sound transmission across the cavity above the critical frequency, but only has a small effect below the critical frequency.

VII. EQUATION OF BEST FIT

The equation for the empirical compliance was obtained by conducting a regression of the natural logarithm of compliance C_M as a function of the natural logarithms of the frequency f , the number of point connections per unit area n or the stud spacing b , and the reduced surface density m_r .

The regression was taken over the low frequency range, 50–400 (or 500) Hz, and the high frequency range, 400 (or 500)–6300 Hz, on the each of four different measurement data sets: Steel studs only, steel studs with resilient channel bars, wood studs with resilient channel bars, and wood studs with resilient channel bars on the both sides. The break between the two frequency ranges was chosen after visual examination of the graphs of the compliance because the compliance appears to decrease as a function of frequency above 400 (or 500) Hz.

Using each side of the equation of best fit as the argument of the exponential function produces the following equations:

$$C_M = A \left(\frac{f}{f_0} \right)^{x_f} \left(\frac{m_r}{m_{r0}} \right)^{x_m} \left(\frac{b}{b_0} \right)^{x_b} \quad (\text{Line connection}), \quad (45)$$

$$C_M = A \left(\frac{f}{f_0} \right)^{x_f} \left(\frac{m_r}{m_{r0}} \right)^{x_m} \left(\frac{n}{n_0} \right)^{x_n} \quad (\text{Point connection}). \quad (46)$$

A is a constant, f is the frequency, f_0 is 1 Hz, m_r is the reduced surface density, m_{r0} is 1 kg/m², b is the distance

TABLE I. Types of gypsum plasterboard used by NRCC and their Young's moduli.

Nominal Type (mm)	Average Young's Modulus (GPa)
AX (16)	2.4
BX (16)	2.18
CX (16)	2.26
A (13)	1.70
B (13)	1.86
Light weight - LB (13)	1.60

TABLE II. Values and confidence limits for the constants in the low and high frequency range with point and line connection models of the measurements of steel stud only construction.

		Constant	Value	95% Upper limit	95% Lower limit
LINE connection	Low frequency	A [1/Pa]	3.22×10^{-4}	4.71×10^{-4}	1.74×10^{-4}
		x_f	-0.55	-0.46	-0.64
		x_m	-1.07	-0.89	-1.26
	High frequency	A [1/Pa]	2.60	3.45	1.97
		x_f	-1.90	-1.87	-1.93
		x_m	-1.29	-1.21	-1.37
POINT connection	Low frequency	A [m/N]	1.88×10^{-4}	3.21×10^{-4}	5.55×10^{-5}
		x_f	-0.28	-0.15	-0.42
		x_m	-1.02	-0.72	-1.31
	High frequency	A [m/N]	0.07	0.09	0.05
		x_f	-1.50	-1.46	-1.54

between the line connections (stud spacing), b_0 is 1 m, n is the number of point connections per unit area calculated from the stud spacing and the resilient channel bar or screw spacing, and n_0 is 1 ($1/\text{m}^2$). The reduced surface density defined in Eq. (36) is used as a way to combine the two wall leaf surface densities because it appears in Eq. (19) for the mass-air-mass resonant frequency. The constant A has units of 1/Pa for line connections and m/N for point connections. The dimensionless exponent constants produced by the regression are x_f , x_m , x_b , and x_n . The regressions of the equivalent translational compliance produced the constants in the equations and their 95% confidence limits.

VIII. RESULTS

The results obtained from the regression analysis are shown in the Tables II–V. There are 16 different equations all together: Two different connection models, four different combinations of studs and/or resilient connections, and two different frequency ranges. The values of the constants and the 95% upper and lower confidence limits of the equivalent translational compliances are shown in the tables. A is statistically significantly different from 1, and x_f , x_m , and x_b are statistically significantly different from 0 at the 95% confidence level when shown in the tables. The tables show only results which are statistically different to zero or one.

There is only one equation where the line of best fit has a statistically significant dependence on the number of point connections per unit area n . This occurs in the low frequency

range for the point connection model of wood studs with resilient channel bars on only one side. No equations showed a statistically significant dependence on the stud spacing b . This is probably because there were only two different stud spacings of 0.406 and 0.61 m. The analysis done by Davy *et al.* (2012) showed that there was no statistically significant dependence on the frequency in the low frequency range for the line connection model of the steel stud only case. The analysis presented in this paper shows statistically significant dependence on frequency in this case.

The equivalent translational compliance is statistically significantly dependent on the reduced mass in 9 of the 16 equations presented in this section.

IX. COMPARISON OF BEST FIT EQUATION

A. Steel stud with resilient channel bar

Figure 1 shows the difference between the steel studs only and steel studs or wooden studs with resilient channel bars on one side of the wall. From the theoretical aspect, the resilient channel bar supplies more compliance, which means the construction with resilient channel bars is expected to obtain higher transmission loss.

The parameters used for calculating the compliance of the typical wall constructions were as follows, and these numbers were the averaged values of the available data used to obtain the regression equations. The reduced surface density for the point connection model was for low frequency 7.95 kg/m^2 , for high frequency 7.44 kg/m^2 , for the line connection low frequency 8.06 kg/m^2 , and for high frequency

TABLE III. Values and confidence limits for the constants in the low and high frequency range with point and line connection models of the construction of steel studs with resilient channel bars on one side of the steel studs.

		Constant	Value	95% Upper limit	95% Lower limit
LINE connection	Low frequency	A [1/Pa]	5.40×10^{-6}	6.50×10^{-6}	4.30×10^{-6}
	High frequency	A [1/Pa]	2.15	4.44	1.04
		x_f	-1.89	-1.82	-1.96
		x_m	-0.98	-0.74	-1.22
POINT connection	Low frequency	A [m/N]	8.18×10^{-6}	1.01×10^{-5}	6.22×10^{-6}
	High frequency	A [m/N]	0.64	1.4	0.29
		x_f	-1.54	-1.46	-1.61
		x_m	-0.95	-1.69	-1.21

TABLE IV. Values and confidence limits for the constants in the low and high frequency range with point and line connection models of the construction of wood studs with resilient channel bars on one side of the wood studs.

		Constant	Value	95% Upper limit	95% Lower limit
LINE connection	Low frequency	A [1/Pa]	2.01×10^{-4}	3.58×10^{-4}	1.13×10^{-4}
		x_f	-0.48	-0.38	-0.58
		x_m	-0.98	-0.76	-1.21
	High frequency	A [1/Pa]	0.22	0.31	0.15
		x_f	-1.54	-1.50	-1.57
		x_m	-1.16	-1.03	-1.28
POINT connection	Low frequency	A [m/N]	5.27×10^{-4}	1.69×10^{-3}	1.65×10^{-4}
		x_f	-0.22	-0.06	-0.38
		x_m	-1.19	-0.80	-1.58
	High frequency	x_n	-0.87	-0.36	-1.38
		A [m/N]	0.06	0.09	0.04
		x_f	-1.16	-1.12	-1.20
		x_m	-1.17	-1.04	-1.29

7.44 kg/m². The number of point connections per square meter used for calculating the compliance for the wood stud with resilient channel bar low frequency was 3.98. The constants in Table II are used for the steel stud only construction and Table III is used for the case of steel studs with resilient channel bars on one side. Table IV is used for the case of wooden studs with resilient channel bars on one side. Equations (45) and (46) are used to obtain the values for compliance for both constructions.

In Fig. 1, the compliance values for the point connection model of the steel studs with resilient channel bars and steel studs only overlap at and above 800 Hz. They are higher than all the other compliances presented. The point connection model of the wood studs with resilient channel bars shows equivalent compliance values to the line connection constructions of the steel stud only, steel stud with resilient channel bars, and wood stud with resilient channel bars. It should be noted that the units for the point and line connection models are different. It can be seen that four of six compliance values are within a narrow range at and above 630 Hz.

B. Comparison with previous studies

Davy *et al.* (2012) compared their regression equation with Vigran (2010) and with Poblet-Puig *et al.* (2009) for the line connection model for 70 mm wide TC steel studs spaced at 600 mm centers with 13 mm gypsum plaster board on each side. These results are now compared with this paper's regression equations for the steel stud only line

connection model. The equation of Vigran (2010), which is a best fit third order polynomial approximation to the numerically calculated equivalent translational compliance data for TC steel studs of Poblet-Puig *et al.* (2009), is as follows:

$$-\log_{10} C_M = 0.6286x^3 - 4.4051x^2 + 10.3323x - 7.0722, \quad (47)$$

where the x is the logarithm to base 10 of the frequency.

The regression equations of Davy *et al.* (2012) are the following:

$$C_M = 9.3 \times 10^{-5} (m_i^{-1.09}) (d^{0.80}) \quad (\text{Low frequency}), \quad (48)$$

$$C_M = 1.74 (f^{-1.81}) (m_i^{-1.81}) (b^{-1.40}) (d^{0.28}) \quad (\text{High frequency}). \quad (49)$$

Figure 2 shows the comparison of the compliance obtained in these previous studies and in this paper. Although the experimental data only go down to 50 Hz, the regression equations are extrapolated down to 16 Hz for comparison with Poblet-Puig's numerically calculated compliances. At the lowest frequency of 16 Hz, the line compliance of 1.24×10^{-5} (1/Pa) obtained in the present study is close to the equation of Vigran (2010) but different from the other two results. At 25 Hz, this paper's equation is close to the numerical result of Poblet-Puig *et al.* (2009) and slowly drops to the value of 2.09×10^{-6} (1/Pa) at 400 Hz, where it

TABLE V. Values and confidence limits for the constants in the low and high frequency range with point and line connection models of the construction of resilient channel bars on both sides of wooden studs.

		Constant	Value	95% Upper limit	95% Lower limit
LINE connection	Low frequency	A [1/Pa]	1.05×10^{-5}	1.95×10^{-5}	2.33×10^{-6}
	High frequency	A [1/Pa]	9.05×10^{-3}	0.17	4.7×10^{-4}
		x_f	-1.25	-0.87	-1.64
POINT connection	Low frequency	A [m/N]	1.79×10^{-5}	3.31×10^{-5}	2.79×10^{-6}
	High frequency	A [m/N]	2.56×10^{-3}	4.78×10^{-2}	1.37×10^{-4}
		x_f	-0.88	-0.49	-1.26

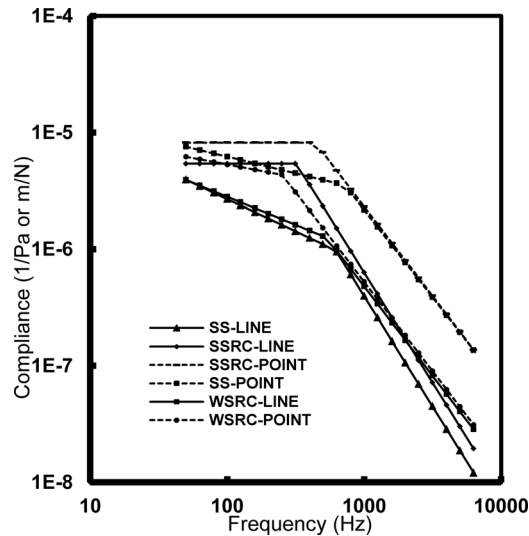


FIG. 1. Comparison of the compliance of steel stud only and steel stud and wooden stud with resilient channel bars for both line connection and point connection models.

is close to the value of 1.92×10^{-6} (1/Pa) given by Davy *et al.* (2012). In the high frequency range, the equation derived in this paper agrees well with the values of Davy *et al.* (2012), Vigran (2010), and Poblet-Puig *et al.* (2009).

C. Thirteen steel studs measurement made by the United States Gypsum Corporation

Davy *et al.* (2012) also presented a comparison of maximum and minimum values of the sound reduction index at each third octave band frequency across 13 measurements on 16 mm gypsum plaster board on each side of 90 mm steel studs with a 610 mm stud spacing commissioned by United States Gypsum Corporation (USG, Chicago, IL), in a

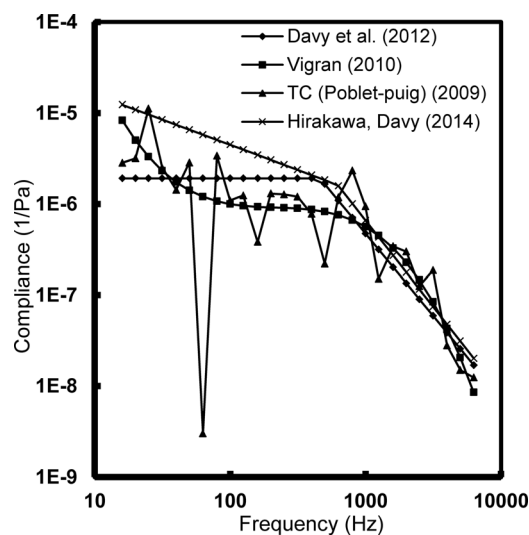


FIG. 2. Comparison of the best fit equations of this paper, Davy *et al.* (2012), and Vigran (2010) for the equivalent translational compliance with the data of Poblet-Puig *et al.* (2009) for 70 mm wide TC steel studs spaced at 600 mm with 13 mm gypsum plasterboard on each side.

number of different laboratories. There was sound absorbing material in the cavities of all these walls.

Figure 3 shows the sound reduction index (dB) predicted by the point and the line connection models and the experimental maximum, average, and minimum sound reduction index at each frequency. Both the line and point connection models predicted values close to the experimental minimum or average in the lower frequency range up to 160 Hz. However, the predictions of both these models relative to the experimental data increase in the frequency range between 160 and 630 Hz where they intercept with the experimental average value between 250 and 315 Hz with a sound reduction index around 42 dB. The interception with the experimental maximum occurred between 400 and 500 Hz with an experimental maximum of 52 dB. The predicted value of the line connection model rises to 64 dB at 1250 Hz where it is 2 dB higher than the experimental maximum. The point connection model was 63 dB at the same frequency.

The critical frequency dip can be seen at 2500 Hz for both connection models and for the experimental average and minimum. The experimental maximum curve has a local minimum at 3150 Hz where it is 1 dB lower than at 2500 Hz where the experimental minimum and average curves have their local minimum.

Both of the prediction values are closer to the experimental maximum for the middle to high frequencies and are close to the experimental minimum at low frequencies. Both connection prediction values show a 2–3 dB higher sound reduction index than the experimental maximum at and above the critical frequency dip.

X. ACCURACY OF REGRESSION

The accuracy of the regressions can be shown by the histogram of differences between the predictions and the

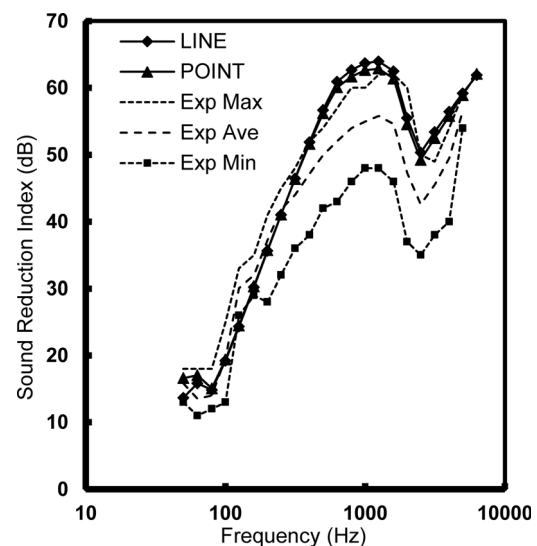


FIG. 3. The maximum, minimum and average experimental values from 13 United States Gypsum Corporation (Chicago, IL) sound insulation measurements on walls with 16 mm gypsum plaster board on each side of 90 mm steel studs spaced at 610 mm compared with the predictions of the point and line models. There was sound absorbing material in the wall cavities.

measurements. If the histogram is distributed with a peak at a difference of zero, the average predicted value is good and the range of the differences is shown by the width of the histogram. If the peak of the distribution occurs for a non-zero difference, the average predicted value has an offset from the average experimental value.

It should be noted that as the resilience of the structural connections between the two wall leaves becomes greater, there is a danger of the experimental sound insulation results being influenced by vibration transmission between the two wall leaves via the frame of the test opening, as can happen in the case of double stud walls (Fausti *et al.*, 1999). Based on comparisons between theory and experiment for double stud walls, it is believed that this is not a problem for the more recent National Research Council Canada (NRCC) data used in this paper. However, some of the USG data appear to have been influenced by flanking transmission.

A. USG (Chicago, IL)

Figure 4 shows the histogram of the differences between theory and experiment for the case of steel stud walls with resilient channel bars on one side which were published by the USG. It is skewed to toward positive differences for both connection models. There is a wide range of differences and the median difference is about 5 dB.

B. NRCC

A total number of eight histograms are shown in Figs. 5 and 6. Four of the eight are indicating peaks at 0 dB difference. In Fig. 5, the resilient channel bars on the both sides of the studs have their peaks at 0 and -2 dB for both connection models. Their two peaks show a degree of inaccuracy that may a result from the small number of this type of measurement present in NRCC's data because there were only

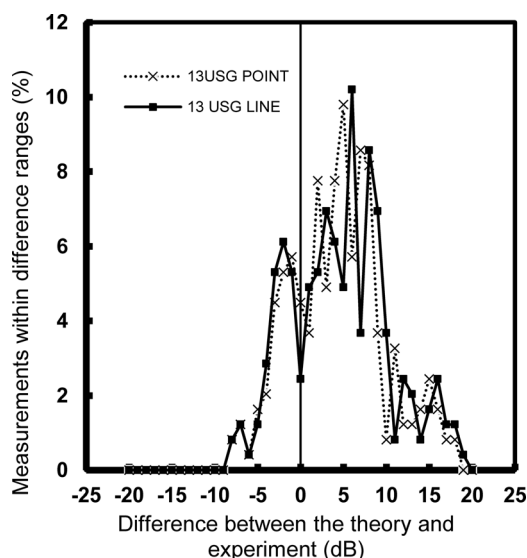


FIG. 4. Histogram of the differences between prediction and experiment for 13 measurements of steel stud only data commissioned by the United States Gypsum Corporation (Chicago, IL) for both the line and point connection models.

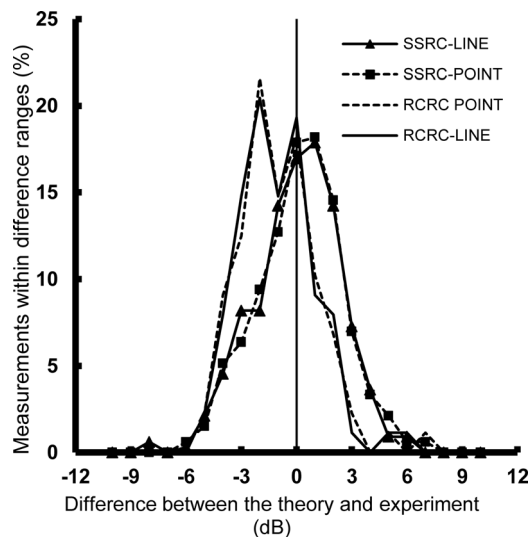


FIG. 5. Histogram of the differences between prediction and experiment for NRCC's measurements on walls constructed using steel stud and resilient channel bars, and resilient channel bars on both sides of wood studs, for both the line and point connection models.

four measurements for this particular type of construction. The steel stud and resilient channel bars construction shows a peak at +1 dB. This means the theory predicts higher than experimental value.

XI. SUMMARY AND DISCUSSION

This paper considered the sound insulation of four different types of cavity stud walls with resilient connections between the two wall leaves using linear and point connection models and with sound absorbing material in their wall cavities. The types of resilient connection studied were steel studs, resilient channel bars on one or both sides of wood studs, and resilient channel bars on one side of steel studs. The values of equivalent translational compliance

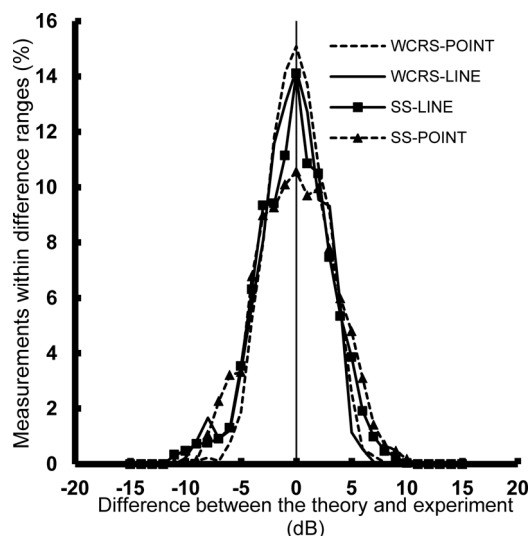


FIG. 6. Histogram of the differences between prediction and experiment for NRCC's measurements on walls constructed using wood stud and resilient channel bars, and steel stud only, for both the line and point connection models.

that made a sound insulation theory agree with experiment were calculated. A linear regression of the logarithm of the compliance against the logarithms of frequency, reduced surface density of the wall leaves, cavity depth and number of point connections or stud spacing, was conducted in both the low frequency range and in the high frequency range. The constants of the regression equations are given in Tables II to V.

Previously, regression equations for the compliance had only been obtained for case of steel studs modeled as line connections (Davy *et al.*, 2012). These previous regression equations were obtained by combining 126 measurements into 28 groups of similar construction types and calculating the compliances needed to make the theory agree with the average experimental sound insulation for each of the 28 groups. In the present paper, these 126 measurements were used individually to calculate the compliance that made theory agree with experiment.

The compliance results for the steel stud case are compared with the previous predictions made by Vigran (2010), Poblet-Puig *et al.* (2009), and Davy *et al.* (2012) in Fig. 2. There was good agreement in the high frequency range with all three previous results, and reasonable agreement in the low frequency range with Vigran (2010) and Poblet-Puig *et al.* (2009).

These results can be used in simple theoretical models for predicting the sound insulation of gypsum plaster board cavity walls with resilient structural connections between their wall leaves.

The maximum, minimum, and average sound reduction index of 13 steel stud measurements produced by USG were compared with the present prediction model in Fig. 3. The results show that the both line and point connection models show great agreement in low frequency range and overlap with the maximum experimental value in the middle to high frequency range. The critical frequency dip is also correctly predicted.

Histograms of the difference between the predicted and experimental values for NRCC's measurements for all construction types and for some of USG's measurements for both connection models were shown in Figs. 4–6. These histograms show the wide range of the differences, but the median values are fairly close to zero except in some cases where there is a noticeable offset from zero difference. These offsets are thought to be due to flanking sound transmission. Some histograms showed many peaks due to the number of differences being less than the number differences needed to create a smooth histogram.

XII. CONCLUSION

Regression equations for the equivalent translational compliance of resilient channel bars on one or both sides of wood studs and on one side of steel studs and for steel studs without resilient channel bars have been derived. These equations have been derived for the connections being modeled both as point and line connections. Previously only regression equations for the line model of steel studs had

been derived. These regression equations can be used in simple theoretical models to predict the sound insulation of gypsum plaster board cavity walls.

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